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MH1200 Linear Algebra I.

The 1st Quiz 2023.

- Please write down your **name** and **matriculation number** at the top. The duration is **20 minutes**. There are **two** questions. A total of **5** marks.
- You may write on every part of this document and also use extra paper if you need. To get full marks please write down all of your working.

Problem 1: (3 marks.)

Consider the following system of linear equations using variables u, v, w, and z.

$$\begin{aligned} -v + 2w - z &= 1 \\ -u + 2v - w &= 1 \\ 2u - v &= -2 \end{aligned}$$

u v w z

(a) Convert this system into an augmented matrix.

(b) Give a sequence of elementary row operations which will transform this matrix into a matrix in row echelon form.

(c) Compute a parametrization for the set of solutions.

Show in full detail how to solve this problem without a calculator, although you may use a calculator to check your solutions if you wish.

a) $\left[\begin{array}{cccc|c} 0 & -1 & 2 & -1 & 1 \\ -1 & 2 & -1 & 0 & 1 \\ 2 & -1 & 0 & 0 & -2 \end{array}\right]$

b) $\left[\begin{array}{cccc|c} 0 & -1 & 2 & -1 & 1 \\ -1 & 2 & -1 & 0 & 1 \\ 2 & -1 & 0 & 0 & -2 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc|c} -1 & 2 & -1 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 2 & -1 & 0 & 0 & -2 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_1} \left[\begin{array}{cccc|c} -1 & 2 & -1 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 3 & -2 & 0 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 + 3R_2} \left[\begin{array}{cccc|c} -1 & 2 & -1 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & 4 & -3 & 3 \end{array}\right]$

c) Set $z = a$, $a \in \mathbb{R}$

$4w - 3z = 3$

$\therefore 4w - 3a = 3$

$4w = 3 + 3a$

$w = \frac{3}{4} + \frac{3}{4}a$

$\therefore -v + 2w - z = 1$

$v = 2w - z - 1$

$= 2(\frac{3}{4} + \frac{3}{4}a) - a - 1$

$= \frac{3}{2} + \frac{3}{2}a - a - 1$

$= \frac{1}{2} + \frac{1}{2}a$

$\therefore -u + 2v - w = 1$

$u = 2v - w - 1$

$= 2(\frac{1}{2} + \frac{1}{2}a) - (\frac{3}{4} + \frac{3}{4}a) - 1$

$= -\frac{3}{4} + \frac{1}{4}a$

$\therefore$ The solution is $\begin{cases} u = -\frac{3}{4} + \frac{1}{4}a \\ v = \frac{1}{2} + \frac{1}{2}a \\ w = \frac{3}{4} + \frac{3}{4}a \\ z = a \end{cases}$, $a \in \mathbb{R}$

Problem 2: (2 marks.)

Consider the following parametrization for a subset $S \subset \mathbb{R}^3$.

$$\begin{cases} x_1 = -1 - 2s - t \\ x_2 = 2 - 3s \\ x_3 = 6 + 2t \end{cases}, s, t \in \mathbb{R}.$$

Express this set using set theory notation, then give a geometric interpretation of this set.

~~It can be expressed~~

$$S = \{ (-1,2,6) + s(-2,-3,0) + t(-1,0,2) \in \mathbb{R}^3,\ s,t \in \mathbb{R} \}$$

This is a plane containing the origin and (-2,-3,0) and (-1,0,2), which is shifted 1 unit in the negative x-direction, 2 units in the positive y-direction, and 6 units in the positive z direction.